

Real-time Target Speech Extraction

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Context of the Problem

01 / HUMAN FOCUS

Innate Human Ability

Humans naturally focus on a single speaker in noisy environments, like a crowded restaurant.



02 / EDGE HARDWARE

Edge AI Integration

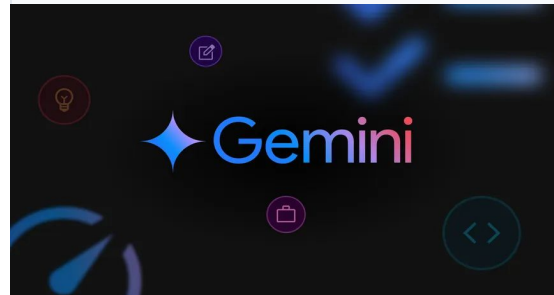
Expanding into everyday devices like smart glasses and hearables to better analyze environmental context.



03 / LIVE EXTRACTION

Real-Time AI Transcription

Applying extraction to live AI tools (e.g., Gemini Live) for robust voice understanding in chaotic environments.



Problem Formulation

PROBLEM FORMULATION

Mixture Audio = Target Speaker + Interfering Speakers + Background Noise

TECHNICAL CHALLENGES

Transforming speech representations and conditioning models with limited target information.

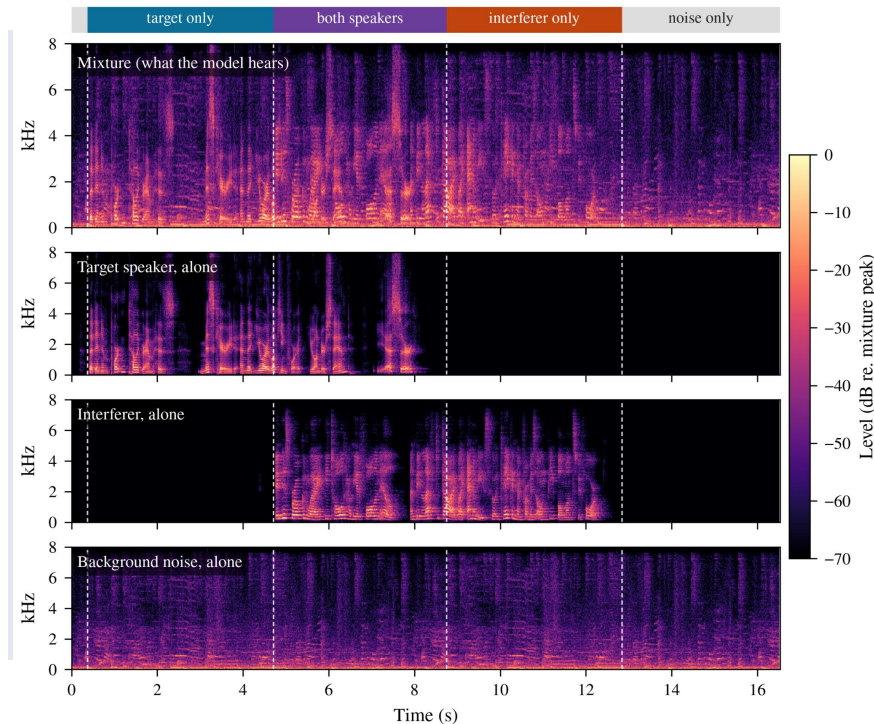
HARDWARE CONSTRAINTS

Balance performance with edge storage limits and strict latency budgets for streaming.

PRIMARY OBJECTIVE

Isolate the target speaker to maximize downstream live AI speech transcription accuracy.

Optimize for AI transcription, remain silent when the target isn't speaking, and suppress noise.



Current Project State



01 // PILLAR

Dataset Construction

Building a dedicated dataset for realistic, noisy multi-speaker use cases.

- Multi-speaker mix scenarios
- Controlled acoustic noise
- Realistic speech benchmarks



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Baseline Signal Extraction

Evaluating architectures that mask mixture audio to establish baselines.

- Target speech isolation
- Time-domain masking
- Baseline performance curves



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Holistic Metric Suite

Developing evaluation metrics tailored for speech transcription performance.

- AI hallucination tracking
- Interference leakage
- Streaming latency budgets
- Perceptual audio quality